Masked Auto-Encoding Spectral–Spatial Transformer for Hyperspectral Image Classification

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***Abstract*— Deep learning has certainly become the dominant trend in hyperspectral (HS) remote sensing (RS) image classification owing to its excellent capabilities to extract highly discriminating spectral–spatial features. In this context, transformer networks have recently shown prominent results in distinguishing even the most subtle spectral differences because of their potential to characterize sequential spectral data. Nonetheless, many complexities affecting HS remote sensing data (e.g., atmospheric effects, thermal noise, quantization noise) may severely undermine such potential since no mode of relieving noisy feature patterns has still been developed within transformer networks. To address the problem, this article presents a novel masked auto-encoding spectral–spatial transformer (MAEST), which gathers two different collaborative branches: 1) a reconstruction path, which dynamically uncovers the most robust encoding features based on a masking auto-encoding strategy, and 2) a classification path, which embeds these features onto a transformer network to classify the data focusing on the features that better reconstruct the input. Unlike other existing models, this novel design pursues to learn refined transformer features considering the aforementioned complexities of the HS remote sensing image domain. The experimental comparison, including several state-of-the-art methods and benchmark datasets, shows the superior results obtained by MAEST. The codes of this article will be available at https://github.com/ibanezfd/MAEST.**

***Index Terms*— Hyperspectral (HS) imaging, mask auto-encoders (MAEs), Vision Transformers (ViTs).**

I. Introduction

OVER the past years, hyperspectral (HS) imaging has certainly emerged as one of the most important technologies for collecting valuable information from the Earth's surface [1]. In this way, numerous remote sensing (RS) applications take full advantage of HS imagery to cope with different challenges and societal needs, such as accurate material identification [2], [3], [4], precision agriculture [5], [6], [7], target detection [8], [9], [10], or environmental management [11], [12], [13] among others. In general, HS instruments are able to acquire data using hundreds of spectral bands to provide detailed information across the electromagnetic spectrum, which logically becomes highly beneficial for fine-grained land-cover classification [14]. In contrast to other data, HS images allow distinguishing even the most subtle spectral differences among similar pixels that certainly help for accurate discrimination of remote sensing image components beyond sensor spatial limits whatsoever [15]. Nonetheless, the substantially higher spectral complexities of the HS image domain often bring important difficulties in successfully exploiting such potential capabilities.

Within the remote sensing field, multiple technologies have been proposed for building effective HS image classifiers based on different features and classification paradigms [16], [17], [18]. Whereas traditional pattern recognition and machine learning approaches are typically constrained by the use of hand-crafted features, deep-learning (DL)-based methods [19], [20], [21] have certainly become the dominant trend owing to the superior capabilities of artificial neural networks (ANNs) to extract highly discriminating spectral–spatial features from HS data. The rapid and successful development of numerous backbone networks for HS remote sensing image classification exemplifies this fact. Since Chen et al. [22] first proposed using stack auto-encoders (AEs) for characterizing HS images, extensive research has been conducted to generate more accurate classifiers. In [23], the authors were able to improve their former results by means of convolutional neural networks (CNNs). Hang et al. [24] took advantage of recurrent neural networks (RNNs) to further exploit the relationships among neighboring HS bands. Zhu et al. [25] designed a generative adversarial network (GAN) to effectively classify HS data based on the use of random input noise. Paoletti et al. [26] also presented a new HS capsule unit (CapsNet) to achieve prominent results while reducing the computational complexity. Besides, Hong et al. [27] created a new graph convolutional network (GCN) to address the problems related to large graphs when managing HS data.

Despite the remarkable results provided by these and other relevant networks, their limited ability to characterize sequential spectral data generally made these models unable to really exploit mid-term and long-term dependencies along the spectral dimension [28]. In fact, it was not until recently that some authors were able to address these limitations using the

Manuscript received 7 July 2022; revised 15 September 2022; accepted 20 October 2022. Date of publication 28 October 2022; date of current version 14 November 2022. This work was supported by the Ministerio de Ciencia e Innovación under Grant PID2021-128794OB-I00. *(Corresponding author: Ruben Fernandez-Beltran.)*

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Digital Object Identifier 10.1109/TGRS.2022.3217892

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so-called tranformers [29]. In [30], Dosovitskiy et al. proposed the Vision Transformer (ViT) for image classification. In brief, ViT makes use of a positional embedding to sequentially process image patches via the well-known self-attention (SA) transformer encoder. More recently, Hong et al. [31] implemented a novel transformer network, termed as SpectralFormer (SF), which went a step further by taking also advantage of spatial contexts and long-term skip connections (SCs). With these improvements, SF has become state-of-the-art in HS remote sensing image classification. However, there are some intrinsic data complexities that still remain unsolved when classifying HS imagery from a transformer-based perspective.

In the context of HS remote sensing imaging, one of the main intricacies is related to the noise present in the acquired data [32]. That is, airborne and spaceborne HS instruments aim at measuring the light reflected by the Earth's surface with an exceptional spectral resolution, but in such a goal, many factors may deteriorate the electromagnetic signal falling onto just a few manometers of spectral width. Without doubt, atmospheric effects together with thermal, quantization, and shot noise are all important factors that may easily affect some pixels of HS remote sensing images [33]. Nonetheless, the most accurate classification technology based on transformers is still unable to take this fact into account inside its own network topology. In other words, although some state-of-the-art models such as ViT and SF are able to exploit subtle spectral discrepancies [30], [31], the robustness of such information may be compromised since no mode of relieving noisy patterns is adopted within the own transformer backbone network. Precisely, this is the research gap that motivates this work.

Unlike regular noise reduction methods that can be considered as the first preprocessing step of the standard HS remote sensing classification pipeline [33], this work pursues a different target based on expanding the concept of transformers to promote a more robust feature learning process inside the own architecture. Following this objective, we rethink the way transformers work with HS data by taking advantage of auto-encoding networks [34]. Specifically, we are interested in the denoising capabilities of mask auto-encoders (MAEs) [35] and how they can be effectively embedded into a new transformer network specifically designed for HS remote sensing image classification. In response, we propose a novel masked auto-encoding spectral–spatial transformer (MAEST) which gathers two different collaborative branches: 1) a reconstruction path and 2) a classification path. On one hand, the reconstruction path adopts an auto-encoding transformer shape which aims at deactivating some pixels from the input HS data while trying to recover such masked data with the objective of identifying the most robust encoding features. On the other hand, the classification path embeds these features onto a transformer network to learn how to classify the complete data while focusing on the features that better reconstruct the input. With this novel design, MAEST pursues to uncover refined encoding features to achieve more accurate predictions and faster convergences with respect to the existing transformers. For validating the proposed approach, a comprehensive experimental comparison is conducted using three benchmark datasets and different state-of-the-art classification models. Therefore, the key contributions of this article can be summarized as follows.

1) We propose a new transformer network (MAEST) specially designed to classify HS remote sensing data. The presented model aims at exploiting auto-encoding denoising capabilities inside the transformer encoder to relieve HS intrinsic noise.
2) We explore the proposed approach performance under multiple configurations to provide insights about the working mechanisms and benefits of MAEST with respect to other state-of-the-art classifiers available in the literature.

The remaining parts of this article are organized as follows. Section II inspects some related works while analyzing their main limitations. Section III describes the proposed classification architecture. Section IV presents and discusses the conducted experiments and studies. Finally, Section V provides the conclusions of this work with an outlook on future research lines.

II. Related Work

A. HS Image Classification

Even though classification of HS imagery is a difficult task due to limited labeled data, the high number of spectral bands, the similarity between spectral signatures between close related classes, and the spectral noise among other concerns, plenty of models have obtained rather good results. Starting from conventional classifiers such as K-nearest neighbors (KNNs) [36], support vector machines (SVMs) [37], or random forest (RF) [38]; to the advances in recent years, thanks to DL. The emergence of CNNs resulted in a relevant improvement in classification accuracy [39]. In [40], Lee and Kwon used a contextual CNN with multiscale convolutional filters and a joint spectral–spatial feature map to improve previous CNN classification results. More recently, Cao et al. [41] proposed a CNN approach with active learning and a Markov random field optimization, reducing the labeling cost and increasing the HS image classification accuracy. Emerged from CNNs, RNNs, which instead of learning from spatial features as 2D-CNNs, use sequential spectral information of the pixel signatures to categorize them also demonstrating the effectiveness in HS image classification [42]. Other DL architectures also showed compelling results. For example, despite their computational cost, GANs [43] have a higher generalization ability than CNNs. Also GCNs, where Hong et al. [27] designed a minibatch GCN (miniGCN) to solve the problems generated by extensive graphs in GCNs for HS imagery. However, these DL classification models usually are unable to properly represent spectral differences between similar pixels and find semantic relationships between pixels or patches globally. In this scene, the transformer architecture seems to be a strong competitor, able to acknowledge these obstacles.

B. Transformers

Transformers are the result of explosion in machine learning due to the use of the SA mechanism, not only in previous

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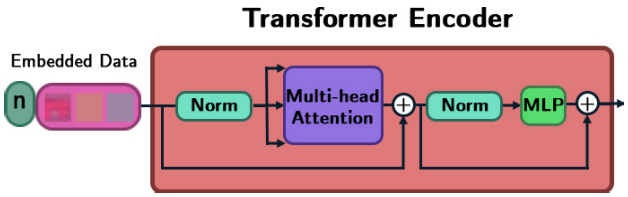


Fig. 1. Transformer encoder from the ViT architecture. The embedded data in position n are sequentially forwarded through normalization layers (Norm), a multihead attention block, and a fully connected layer [multilayer perceptron (MLP)] with residual SCs.

existing models as CNNs but also in entirely new architectures. First in natural language processing (NLP) [44], and now in several image processing fields with the introduction of ViT [30]. In contrast to classic CNNs [41], GANs [43], or GCNs [27], transformers analyze and process the data in a sequential way. In ViT, the input data are forwarded through several transformer encoders. The transformer encoder architecture is shown in Fig. 1. First, image patches are embedded with positional information and fed to a multihead attention block. In this block, the most relevant features of an embedded patch and its global relationships with others are enhanced. As opposed to CNNs and GANs, thanks to the model knowledge of the relative position of all the input sequences, transformers do not rely on past hidden states and recurrence to capture and model middle- and long-range dependencies as well. This is especially important in HS imagery because there is a limitation in the classification of spectral pixel signatures using spatial information (CNNs), as relationships between spectral bands inside the pixel and might contain more information in a similar fashion when there are a number of categories of materials with similar spectral absorption. In the case of RNNs, even though they can learn sequentially ordered spectral information, this also obscures global long-range dependencies between pixels and spectral signatures. Besides, RNNs are not very efficient in real-world applications. However, directly applying ViT to HS image classification tasks also has some drawbacks, including the loss of local and semantic information. In this context, the SF [31] was able to use spectral information from groups of bands with SCs to maintain the semantic features. Yet, other issues such as high spectral noise in HS imagery significantly affect sequential learning. Therefore, a higher generalization capability and robustness are required in HS image classification. For this kind of feature extraction, in image processing, it is common to use AEs [45].

C. Auto-Encoding

The technique of sequentially using an encoder, which generates a latent representation space from the input, and a decoder that uses this representation to reconstruct the input information is called auto-encoding [46]. There are several uses for AEs such as data compression [47], unmixing [48], or data generation [49]. AEs have also been applied in the remote sensing field as well, in anomaly detection [50] or even feature extraction for classification [45], [51]. Another common use for AEs is denoising [52], [53]. Denoising AEs (DAEs) learn to reconstruct the original data from a corrupted input. Some of these DAEs use masked pixels [35] as corruption in the input. Later, DAEs were used for robust feature extraction [53], and then from this idea, MAEs for image reconstruction and pretraining were designed.

D. Masked Image Reconstruction

The masked image reconstruction methods use corrupted images as inputs, learn their latent representations, and perform a reconstruction of the uncorrupted image. Starting from DAEs [53] and thanks to the success of transformers in NLP [44], from the ViT [30] masked patch reconstruction in a self-supervised fashion for feature extraction has been widely used. A great example is bidirectional encoder representation from image transformers (BEiT) [54], which uses masked tokenized patches of the image to predict it, learning to extract general and robust representations as a pretraining for ViT. In [35] Chen et al. used masked and autoregressive pixel prediction to improve the learned deep representations as well. He et al. [55] used a self-supervised asymmetric transformer MAE to reconstruct masked patches from a limited amount of unmasked data.

E. Self-Supervised Learning

In self-supervised learning, unlabeled or pseudolabeled data can be exploited in a supervised manner. Self-supervised learning as a semisupervision technique has been explored for a long time [56], [57], and later in many fields [58], [59] [60]. Though pseudolabeling has proven to be an effective self-supervised learning method for CNNs [61], it still requires generating fabricated labels. In addition, self-supervised learning in AEs allows to use completely unlabeled data to learn deep representations. In HS imagery, this is critical as a result of the small percent of labeled data. In response, inspired by the MAE [55] and the SF [31], in this work we propose the MAEST, a two-branched model with a self-supervised spectral adapted transformer with a masked reconstruction AE to learn generalizing noise-free representations from HS imagery to later classify them.

III. Methodology

The purpose of this work is to introduce the masked autoencoder spectral–spatial transformer (MAEST), a new general HS classification pipeline based on the success of MAE and ViT architectures in computer vision tasks, focused on learning robust features from highly noisy spectral information. The general architecture is organized into two connected pipelines: the reconstruction pipeline and the classification pipeline. The general strategy of MAEST is described in Section III-A. These pipelines are composed of three main blocks: the reconstruction encoder (RE) in Section III-B and the reconstruction decoder (RD) in Section III-C for the first one, and the classification encoder (CE) in Section III-D in the second. Each of these blocks is designed to specifically learn information from HS image data through specialized modules. A general scheme of the complete MAEST is given in Fig. 2.

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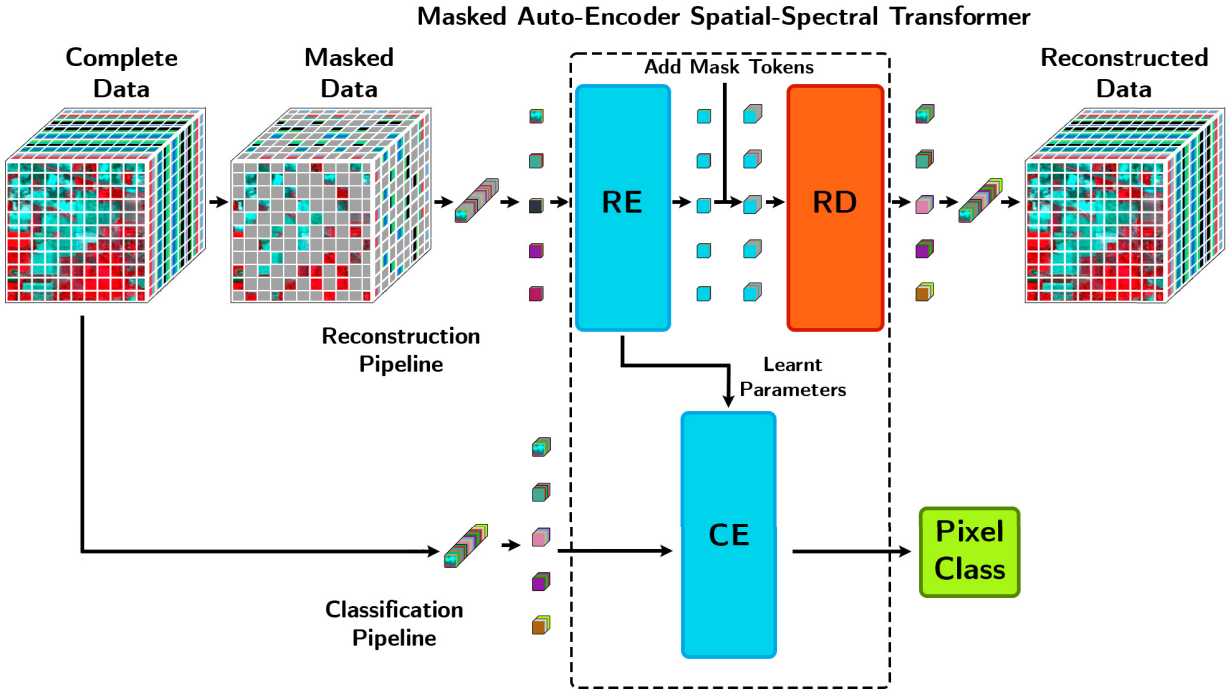


Fig. 2. General scheme of MAEST, including reconstruction and classification pipelines.

A. MAEST Strategy

The MAEST HS image classification approach consists of two interconnected pipelines. In the first one, a masked self-supervised AE learns to reconstruct the original HS image, given a partial observation. This AE is made of two blocks: the RE, an encoder which extracts a latent representation for unmasked segments of the spectral signature of each pixel, and the RD, a decoder that reconstructs the masked data from this latent representation. During this process, RE learns to extract robust spectral–spatial features from the masked HS image, eliminating spectral noise and redundancy. Masking a high percentage of the input data completely changes the easily solved reconstruction task to a challenging process, where robust and generalizing features to represent the small subset of available information are required. Thus, masking the HS image data in random spatial pixels or patches, and groups of spectral bands preventing a possible location bias toward image reconstruction. In addition, it allows to use the encoder efficiently, as it only uses a percentage of the training data. Also, to train RE and RD, we use the unlabeled training data in a self-supervised way. To this end, a simple mean squared error reconstruction loss $\mathcal{L}_r$ ((1)) is used

$$\mathcal{L}_r = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{1}$$

where N is the number of sampled values, and $\hat{y}_i$ and y_i are the ith output and expected ith output, respectively.

The second pipeline is responsible for supervised classification. In this branch, the complete labeled training data are used as input by a single block, the CE. This encoder exploits the robust feature extraction learned parameters in the reconstruction pipeline to categorize pixels after a short fine-tuning of the encoder and the classification layer. For this training, we chose the cross entropy loss $\mathcal{L}_c$, which is defined as

$$\mathcal{L}_c = -\sum_{c=1}^{C} y_c \log \hat{y}_c \tag{2}$$

where $\hat{y}_c$ is the network output, and y_c is the expected class, where c and i are the class indices ranging from $[0, C)$ for a C number of classes.

B. Reconstruction Encoder (RE)

The RE block has the objective of learning robust spectral–spatial features from the masked HS image, to then feed these features to RD and provide guided weights to CE. The RE and all the blocks composing the complete MAEST are based on the ViT backbone architecture with some adaptations for HS images. Due to HS images containing hundreds of spectral bands, the information along the electromagnetic spectrum from HS images suggests a more continuous behavior than red green blue (RGB) images from the visible spectrum. Besides, the information to classify many interesting categories in RS lies in intervals or groups of spectral bands, as characteristic absorption wavelengths. To take advantage of the spectral information, instead of using image patches as input, i.e., ViT, in [31] Hong et al. introduced groupwise spectral embedding (GSE), which uses groups of spectral bands along a spatial patch or pixel as input. Being $\mathbf{X} \in \mathbb{R}^{(H\times W\times M)}$ an HS image of height H, width W, and M spectral bands, and $\mathbf{p} = [p_1, p_2, \ldots, p_M] \in \mathbb{R}^{(1\times M)}$ a single pixel

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along the spectral dimension of $\mathbf{X}$, the embedded features $\mathbf{E} \in \mathbb{R}^{(D\times M)}$ are obtained by the standard ViT

$$\mathbf{E} = \mathbf{wp} \tag{3}$$

where $\mathbf{w} \in \mathbb{R}^{(D\times 1)}$ is the linear projection used for every band. In the case of GSE, the embedded features are generated from local spectral groups which are defined as $\mathbf{p}_G = f(\mathbf{p}) = [p_1, \ldots, p_Q, \ldots, p_M]$, where the function $f()$ generates the overlapping groups of bands and $\mathbf{p}_G \in \mathbb{R}^{(N\times M)}$. With N being the number of bands for each group, $\mathbf{p}^Q = [p_{Q-(N/2)}, \ldots, p_Q, \ldots, p_{Q+(N/2)}]^T \in \mathbb{R}^{(N\times 1)}$. Equation (4) shows the definition of the embedded features generated by GSE

$$\mathbf{E}_G = \mathbf{W}\mathbf{p}_G \tag{4}$$

where $\mathbf{W} \in \mathbb{R}^{(D\times N)}$ are the grouped linear projection for each group $\mathbf{p}_G$ generating the embedded dimensions $\mathbf{E}_G \in \mathbb{R}^{(D\times M)}$. However, in RE we use a modified version of GSE, the masked groupwise spectral embedding (MGSE). Instead of using as input for the encoders the complete set of embedded feature groups $\mathbf{E}$, once the groups are projected we add a learnable positional encoding to the feature groups and randomly mask a K percentage of them. Masking is done after projection to exploit $\mathbf{W}$ weights in CE as well. Due to the electromagnetic spectrum of an HS image containing sampled bands each few nanometers, even though this provides more information of the properties of specific materials, it also includes noise and irrelevant information. RE uses MGSE to learn features that are robust to this noise, improving the classification results and performance. The learnable absolute positional embedding also retains the information of the location of each group in the complete pixel spectrum, even after masking a high percentage of data. Then, MGSE embedded features are modeled as

$$\mathbf{E}_M = g(\mathbf{s}_G + \mathbf{W}\mathbf{p}_G). \tag{5}$$

In this case, $\mathbf{s}_G \in \mathbb{R}^{(D\times M)}$ are the absolute positional embedding, and $g()$ is a function which masks a k percentage of the groups and withdraws them from the input groups. The resulting unmasked embedded dimensions $\mathbf{E}_M \in \mathbb{R}^{(D\times K)}$ where $K = (k-1)\%N$ are used as input. To take advantage of spatial information, MAEST can also take spatial patches $\mathbf{P} \in \mathbb{R}^{(h\times w\times K)}$ with the patch height and width dimensions being h and w, respectively, are flattened, and the embedded features are calculated similarly with each group being of size $\mathbf{p}_Q \in \mathbb{R}^{(Nhw\times 1)}$. In this way, not only the spectral but also the spatial context for the pixel are known in the learning process. For simplicity, the pixel example is used in Sections III-C and III-D. Later, the embedded information is forwarded to a sequence of classic transformer encoders which generate the output reconstruction latent space $\mathbf{L}_E \in \mathbb{R}^{(D_E\times K)}$ of D_E dimensions. The use of only a small percentage of the spectral groups makes the learning of features a low and efficient computation process. The complete scheme of RE is portrayed in Fig. 3.

C. Reconstruction Decoder (RD)

The RD block's goal is to use the latent space features extracted by RE to reconstruct the masked spectral–spatial information of the original HS image. Once RE has generated the reconstruction latent space $\mathbf{L}_E$, a fully connected layer is used as a bridge to RD input size latent space $\mathbf{L}_D \in \mathbb{R}^{(D_D\times K)}$ of D_D dimensions. Then, in the position of the missing masked information, masked tokens are placed. After masked tokens' incorporation, a position embedding is added to the whole new set $\mathbf{L}_M \in \mathbb{R}^{(D_D\times N)}$, therefore including the latent space representations $\mathbf{L}_D$, the masked tokens, and the absolute location information of both. Next, the complete set $\mathbf{L}_M$ is used as input by a sequence of transformer decoders. Finally, a fully connected layer is used as a reconstruction layer to predict only the masked pixels' groups. We include a visual example of the masked reconstruction results in Fig. 4.

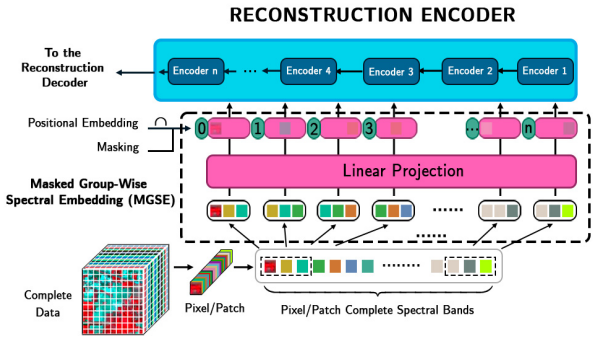


Fig. 3. Detailed diagram of RE.

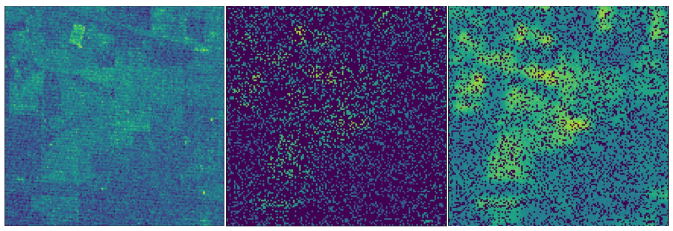


Fig. 4. False color visualization of the reconstruction done by the patch MAEST for the first band for the Indian Pines dataset. (a) Original. (b) Masked. (c) Reconstructed.

D. Classification Encoder (CE)

The last block of MAEST, CE is designed for the HS image classification task. The CE takes advantage of the robust spectral feature extraction learned during HS image reconstruction, and after a short fine-tuning of the inherited parameters it is able to accurately classify HS image categories. In this case, the input of CE is the complete and unmasked training data. In contrast to RE, the original GSE method is used to create the spectral groups of pixels p_G without masking. From $\mathbf{p}_G$ groups and $\mathbf{W}$ projections, the embedded features $\mathbf{E}$ are obtained as in (4). Then, a learnable classification extra token is included in the embedded features, now $\mathbf{E}_C \in \mathbb{R}^{(D\times M+1)}$. This extra classification token is used to assign a category to the complete pixel spectrum. Besides, a positional embedding is included in each of the embedded features and these are forwarded to the sequential modules of encoders in the same way as RE.

However, in CE, an SC mechanism defined as cross-layer adaptative fusion (CAF) [31] is used. This technique was

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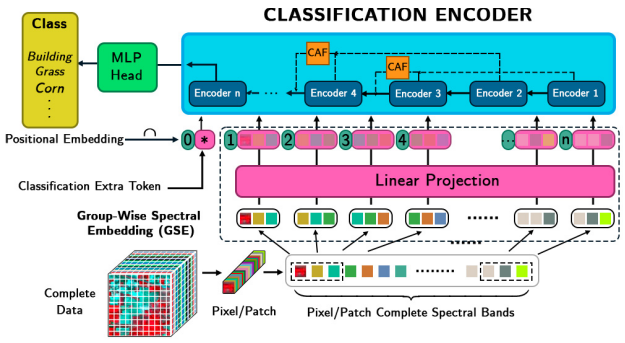


Fig. 5. Detailed diagram of the classification encoder.

inspired by the improvement seen in networks such as ResNet [62] and UNet [63] in computer vision tasks using short-range and long-range SC, respectively. In the case of CAF, a middle-range SC is used to avoid having a huge difference between features (long-range) and a high computational cost (short-range). Also, in HS image classification, it is usual to design shallow network architectures due to the limited labeled training and test data, making long-range SC of several layers unachievable. Specifically, only one encoder is skipped in each connection. The CAF process can be expressed as

$$\hat{z}^{(i)} \leftarrow \ddot{\mathbf{w}} \begin{bmatrix} z^{(i)} \\ z^{(i-2)} \end{bmatrix} \tag{6}$$

where $z^{(i)} \in \mathbb{R}^{(1\times D_z)}$ and $z^{(i-2)} \in \mathbb{R}^{(1\times D_z)}$ are the (i)th and $(i-2)$th feature layers of D_z dimensions, respectively, used as input, and $\ddot{\mathbf{w}} \in \mathbb{R}^{(1\times 2)}$ are the learnable adaptive fusion weights used to obtain the new cross-layer representations $\hat{z}$. Finally, an MLP head is used as a classification layer. A detailed diagram of CE is shown in Fig. 5.

IV. Experiments

In this section, extensive experimentation is conducted to measure the effectiveness of MAEST pretraining for HS classification compared with baseline and state-of-the-art methods. First, three popular HS datasets used through the experimentation are described in Section IV-A, followed by the experimental setting and implementation details in Section IV-B. Then, we evaluate its quantitative and qualitative results for the HS classification task in Section IV-C. Finally, we analyze different configurations and training situations for our model in Section IV-D.

A. Datasets' Description

To study the performance of the proposed MAEST, we use three well-known HS datasets, the Indian Pines dataset, the Pavia University dataset, and the Houston2013 dataset.

1) Indian Pines Dataset: The Indian Pines dataset is composed of 145 × 145 pixels sampled with a resolution of 20 m ground sampling distance (GSD) and 220 spectral bands from 400 to 2500-nm wavelength. The dataset was gathered by the airborne visible/infrared imaging spectrometer (AVIRIS) HS sensor in 1992 over the Purdue University Agronomy farm northwest of West Lafayette and the surrounding area. After removing the water absorption bands, the number of spectral bands is reduced to 200, specifically bands [1–103], [109–149], and [164–219] are retained. The Indian Pines dataset is labeled into 16 main classes. We include details of the training and testing data samples for every class in Table I. A visualization of the training and testing data is also included in Fig. 6.

TABLE I

Land-Cover Classes of the Indian Pines Dataset, With the Standard Training and Testing Sets for Each Class

Class No.	Class Name	Training	Testing
1	Corn Notill	50	1484
2	Corn Mintill	50	784
3	Corn	50	184
4	Grass Pasture	50	447
5	Grass Trees	50	697
6	Hay Windrowed	50	439
7	Soybean Notill	50	918
8	Soybean Mintill	50	2418
9	Soybean Clean	50	564
10	wheat	50	162
11	Woods	50	1244
12	Building Grass Trees Drives	50	330
13	Stone Steel Towers	50	45
14	Alfalfa	15	39
15	Grass Pasture Mowed	15	11
16	Oats	15	5
	Total	695	9671

TABLE II

Land-Cover Classes of the Pavia University Dataset, With the Standard Training and Testing Sets for Each Class

Class No.	Class Name	Training	Testing
1	Asphalt	548	6304
2	Meadows	540	18146
3	Gravel	592	1815
4	Trees	524	2912
5	Metal Sheets	265	1113
6	Bare Soil	532	4572
7	Bitumen	375	981
8	Bricks	514	3364
9	Shadows	231	795
	Total	3921	40002

2) Pavia University Dataset: The Pavia University dataset contains an image of 610 × 340 pixels with 1.3-m GSD and 103 spectral bands from 430 to 860 nm. This image was obtained by the reflective optics system imaging spectrometer (ROSIS) sensor over Pavia University and its surroundings in Pavia, Italy. The Pavia University dataset is classified into nine different land-cover classes. We include details of the training and testing data samples for every class in Table II. A visualization of the training and testing data is also included in Fig. 7.

3) Houston2013 Dataset: The Houston2013 dataset consists of a 349 × 1905 pixels HS image with 144 bands ranging from 364 to 1046 nm. This image contains the area of the campus of the Houston University and its surroundings in Texas, USA, acquired by innovation, technology, research,

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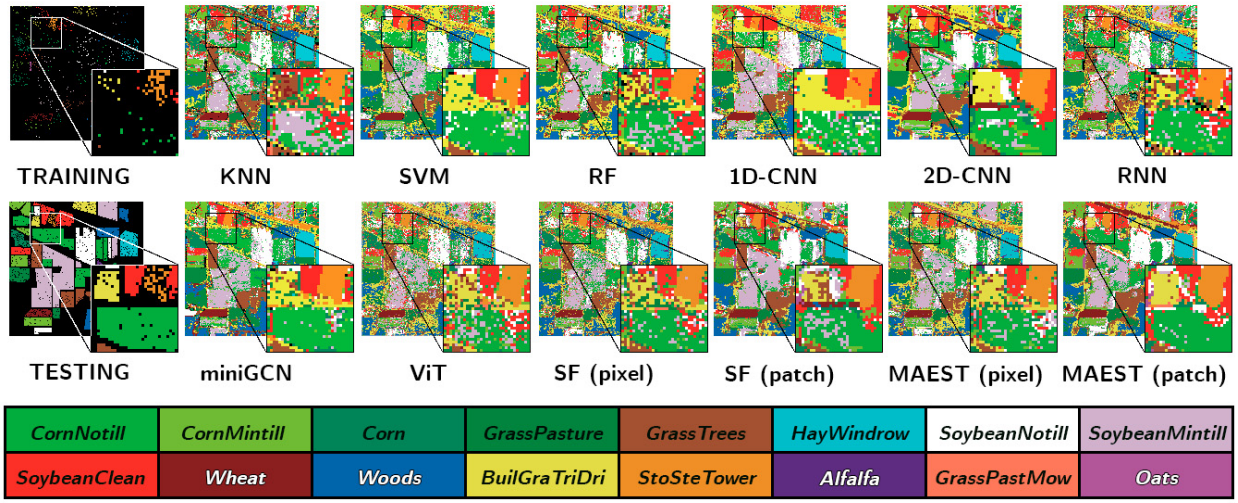


Fig. 6. Spatial distribution of the training and testing sets, and the classification maps generated by different models on the Indian Pines dataset.

TABLE III
LAND-COVER CLASSES OF THE HOUSTON2013 DATASET, WITH THE STANDARD TRAINING AND TESTING SETS FOR EACH CLASS

Class No.	Class Name	Training	Testing
1	Healthy Grass	198	1053
2	Stressed Grass	190	1064
3	Synthetic Grass	192	505
4	Tree	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking Lot1	192	1041
13	Parking Lot2	184	285
14	Tennis Court	181	247
15	Running Track	187	473
	Total	2832	12197

excellence, and service (ITRES) compact airborne spectrographic imager (CASI)-1500 sensor. For our experimentation, we used a cloudless version of the Houston2013 dataset [31]. In this version, the image has been processed to remove occlusions and restore lost data. The Houston2013 dataset has 15 different land-cover and land-use classes. We include details of the training and testing data samples for every class in Table III. A visualization of the training and testing data is also included in Fig. 8.

B. Experimental Settings

1) Implementation: The implementation of the MAEST model and HS classification were designed and executed using PyTorch in Python 3.6 on a Ubuntu 16.04 x64 machine with Intel(R) Core(TM) i7-6850 K processor with 110-GB RAM with an NVIDIA GeForce GTX 1080 Ti 11 GB graphics processing unit (GPU). Due to computational limitations, we executed the masked self-supervised reconstruction and supervised classification separately. Specifically, for classification training, we used a batch size of 64 with the Adam optimizer. A learning rate of $5e^{-4}$ decaying with a γ factor of 0.9 every 1/10 of the epochs was used, which were set to 1000. Nevertheless, both the proposed MAEST and SF converge in much fewer epochs. For MAEST, 300 epochs in the Indian Pines dataset, 400 epochs in Pavia University, and as fast as 260 for the Houston dataset. In the case of SF, 300 epochs as well for the Indian Pines dataset and 500 epochs for the other two datasets.

2) Compared Methods: In the experimentation, we compare the proposed MAEST with many representative classification standard methods, classic backbones, and state-of-the-art models. We divided these classifiers into four main groups: conventional classifiers, classic backbone networks, transformer-based models, and the proposed MAEST. Below, standard parameter selection for each of the classifiers is included.

1) *Conventional Classifiers:* In this group, we included KNN [36] SVM [37], and RF [38]. For KNN, we used a number of nearest neighbors of 10, which is the main factor. The SVM was implemented through the libsvm toolbox3, using the radial basis function kernel. This kernel is defined into two main parameters σ and λ. These parameters are optimized using a fivefold cross-validation in training ranging from $\sigma = [2-3, 2-2, \ldots, 24]$ and $\lambda = [10-2, 10-1, \ldots, 104]$. For the RF main parameter, 200 decision trees were selected.
2) *Classic Backbone Networks:* In the second group, models based on CNNs are included, 1D-CNN [39], 2D-CNN [41], an RNN [42], and miniGCN [27]. The 1D-CNN consists of a set of 128 1D convolutional filters, a batch normalization (BN) layer, and a rectified linear unit (ReLU) activation layer with a softmax for classification purpose. The 2D-CNN contains three blocks made of a 2-D convolutional layer, a BN layer, max-pooling, and finally a ReLU activation layer. The parameters for the convolutional layers are a kernel size of (3×3) for the two first layers and (1×1) for the third. The number of filters for each layer is

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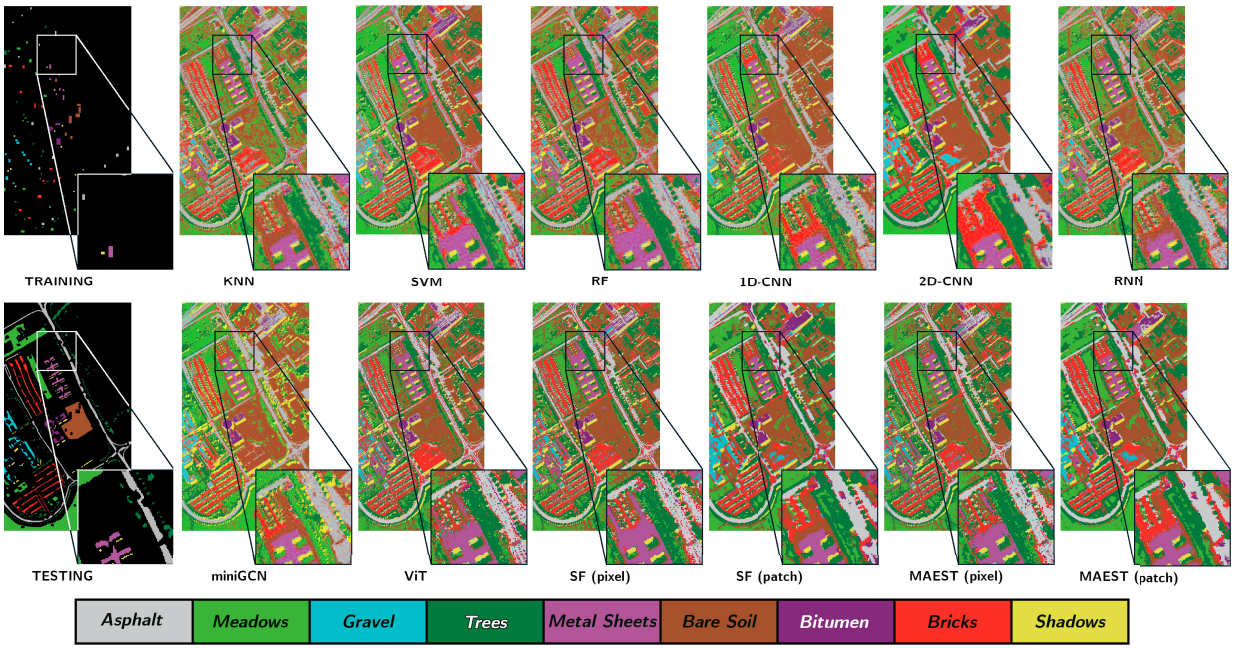


Fig. 7. Spatial distribution of the training and testing sets, and the classification maps generated by different models on the Pavia dataset.

TABLE IV

QUANTITATIVE PERFORMANCE OF DIFFERENT CLASSIFICATION METHODS IN TERMS OF OA, AA, AND κ AS WELL AS THE ACCURACIES FOR EACH CLASS ON THE INDIAN PINES DATASET. BEST RESULTS ARE SHOWN IN BOLD

Class	Conventional Classifiers			Classic Backbone Networks				Transformers			MAEST	
	KNN	SVM	RF	1D-CNN	2D-CNN	RNN	miniGCN	ViT	SF(pixel)	SF(patch)	pixel	patch
1	45.45	67.34	57.80	47.83	65.90	69.00	72.54	41.55	75.07	68.35	68.64	**78.97**
2	46.94	67.86	56.51	42.35	76.66	58.93	55.99	63.39	62.76	79.72	76.91	**92.73**
3	77.72	93.48	80.98	60.87	92.39	77.17	92.93	75.54	92.39	95.11	97.28	**98.91**
4	84.56	94.63	85.68	89.49	93.96	82.33	92.93	68.68	93.29	95.52	93.51	**96.20**
5	80.06	88.52	79.34	92.40	87.23	67.72	**94.98**	81.77	84.79	79.63	87.52	89.10
6	97.49	94.76	95.44	97.04	97.27	89.07	**98.63**	83.60	89.29	99.32	89.98	98.41
7	64.81	73.86	77.56	59.69	77.23	69.06	64.71	71.68	80.50	81.70	84.64	**84.86**
8	48.68	52.07	58.85	65.38	57.03	63.56	68.78	72.08	60.26	66.96	69.44	**73.44**
9	44.33	72.70	62.23	**93.44**	72.87	65.07	69.33	52.84	76.24	65.25	74.11	69.50
10	96.30	98.77	95.06	99.38	**100.00**	95.06	98.77	95.06	98.15	99.38	98.15	**100.00**
11	74.28	86.17	88.75	84.00	92.85	88.67	87.78	92.77	91.00	**93.73**	84.97	92.12
12	15.45	71.82	54.24	86.06	88.18	50.00	50.00	55.45	62.73	84.85	76.36	**91.51**
13	91.11	95.56	97.78	91.11	**100.00**	97.78	**100.00**	97.78	**100.00**	95.56	**100.00**	**100.00**
14	33.33	82.05	56.41	84.62	84.62	66.67	48.72	**89.74**	**89.74**	76.92	94.87	**89.74**
15	81.82	90.91	81.82	**100.00**	**100.00**	81.82	72.73	90.90	90.90	**100.00**	90.90	**100.00**
16	40.00	**100.00**	**100.00**	80.00	**100.00**	**100.00**	80.00	60.00	**100.00**	**100.00**	**100.00**	**100.00**
OA (%)	59.17	72.36	69.80	70.43	75.89	70.66	75.11	69.66	75.69	78.55	78.52	**84.15**
AA (%)	63.90	83.16	76.78	79.60	86.64	76.37	78.08	74.55	84.19	86.37	86.71	**90.97**
κ (%)	0.5395	0.6888	0.6591	0.6642	0.7281	0.6673	0.7164	0.6528	0.7250	0.7566	0.7567	**0.820**

32, 64, and 128. Finally, a softmax is added as well. In the RNN case, two recurrent layers with gated recurrent units of 128 neurons were used. For miniGCN, a block containing a BN layer, a 128 neuron graph convolutional layer, and a ReLU layer were used.

3) *Transformers:* In this third group, we included the original ViT [30] and SF [31]. For the original ViT network architecture, five encoder blocks were used. Each of them contain 4 SA heads, an MLP with 8 hidden dimensions, and a Gaussian error linear unit (GELU) activation layer. A 10% dropout layer is used after positional embedding and MLPs. The SF is used with five encoders using the same parameters with CAF skipping 1 encoder. For patchwise SF, a patch size of 7×7 is used along with a weight decay of $5e^{-3}$.

4) *MAEST:* For RE and RD, we used the same parameters as ViT, and for CE the SF parameters, including CAF and patchwise sizes. The masking percentage for RE was fixed to a random 75% of the samples for self-supervised reconstruction. Due to computational limitations, first the masked reconstruction is performed, and later classification using the learned weights from the reconstruction task.

C. Classification Results

1) Quantitative Results: The classification performances of every presented method and the proposed MAEST are shown in three tables. The Indian Pines dataset results are shown in Table IV, the Pavia University dataset in Table V, and Table VI the Houston2013 dataset results. In the tables, three general

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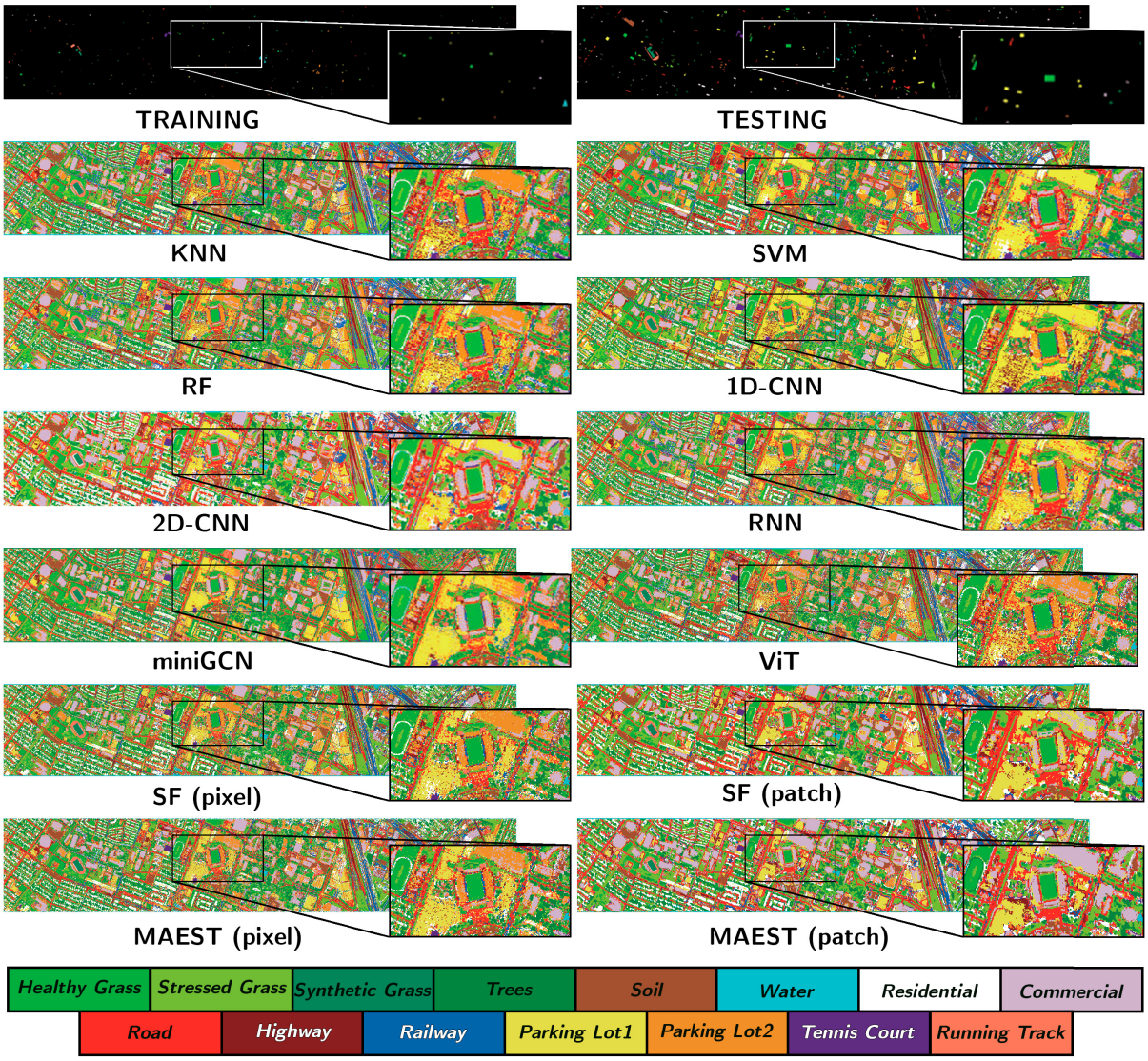


Fig. 8. Spatial distribution of the training and testing sets, and the classification maps generated by different models on the Houston dataset.

TABLE V

QUANTITATIVE PERFORMANCE OF DIFFERENT CLASSIFICATION METHODS IN TERMS OF OA, AA, AND κ AS WELL AS THE ACCURACIES FOR EACH CLASS ON THE PAVIA UNIVERSITY DATASET. BEST RESULTS ARE SHOWN IN BOLD

Class	Conventional Classifiers			Classic Backbone Networks				Transformers			MAEST	
	KNN	SVM	RF	1D-CNN	2D-CNN	RNN	miniGCN	ViT	SF(pixel)	SF(patch)	pixel	patch
1	73.86	74.22	79.81	88.90	80.98	84.01	**96.35**	80.17	76.92	79.77	85.45	79.85
2	64.31	52.79	54.90	58.81	81.70	66.95	89.43	63.24	68.59	86.32	77.24	**96.60**
3	55.10	65.45	46.34	73.11	67.99	58.46	**87.01**	65.45	62.37	72.12	68.60	67.66
4	94.95	97.42	**98.73**	82.07	97.36	97.70	94.26	93.92	97.77	97.87	97.49	96.84
5	99.19	99.46	99.01	99.46	99.64	99.10	**99.82**	99.19	96.14	99.73	99.28	99.73
6	65.16	93.48	75.94	**97.92**	97.56	83.18	43.12	90.75	94.38	79.31	90.94	82.70
7	84.30	87.87	78.70	88.07	82.47	83.08	90.96	87.77	90.11	80.83	89.09	**96.23**
8	84.10	89.39	90.22	88.14	**97.62**	89.63	77.42	89.77	92.39	95.65	91.20	95.96
9	98.36	**99.87**	97.99	99.87	95.60	96.48	87.27	99.12	99.62	90.56	99.50	94.47
OA (%)	70.53	70.82	69.67	75.50	86.05	77.13	79.79	75.93	78.60	85.79	83.70	**91.06**
AA (%)	79.68	84.44	80.18	86.26	88.99	84.29	85.07	85.49	86.48	86.91	88.76	**90.00**
κ (%)	0.6268	0.6423	0.6237	0.6948	0.8187	0.7101	0.7367	0.6980	0.7297	0.8123	0.7899	**0.8794**

different metrics are evaluated: the overall accuracy (OA), the average accuracy (AA), and the Kappa coefficient (κ). The accuracy results for each class are shown as well, with the best results shown in bold font for all the metrics. In general, there is a trend in all the tables where the worst results are in the conventional classifiers, followed by the classic backbone

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TABLE VI

QUANTITATIVE PERFORMANCE OF DIFFERENT CLASSIFICATION METHODS IN TERMS OF OA, AA, AND κ AS WELL AS THE ACCURACIES FOR EACH CLASS ON THE HOUSTON2013 DATASET. BEST RESULTS ARE SHOWN IN BOLD

Class	Conventional Classifiers			Classic Backbone Networks				Transformers			MAEST	
	KNN	SVM	RF	1D-CNN	2D-CNN	RNN	miniGCN	ViT	SF(pixel)	SF(patch)	pixel	patch
1	83.19	83.00	83.38	87.27	85.09	82.34	**98.39**	83.19	80.82	84.52	85.00	82.90
2	95.68	98.40	98.40	98.21	**99.91**	94.27	92.11	94.64	99.15	99.34	98.40	99.44
3	99.41	99.60	98.02	**100.00**	77.23	99.60	99.60	99.60	99.80	89.31	99.80	94.65
4	97.92	**98.48**	97.54	92.99	97.73	97.54	96.78	94.43	95.74	98.30	96.50	96.02
5	96.12	97.82	96.40	97.35	99.53	93.28	97.73	97.92	98.01	**100.00**	98.39	99.24
6	92.31	90.91	**97.20**	95.10	92.31	95.10	95.10	95.10	95.80	93.71	95.10	93.01
7	80.88	90.39	82.09	77.33	**92.16**	83.77	57.28	82.84	83.12	86.85	88.53	89.93
8	48.62	40.46	40.65	51.38	79.39	56.03	68.09	53.94	54.70	80.82	57.64	**82.15**
9	72.05	41.93	69.78	27.95	**86.31**	72.14	53.92	58.73	67.23	76.11	69.59	77.15
10	53.19	62.64	57.63	**90.83**	43.73	84.17	77.41	58.30	87.45	77.61	89.29	87.55
11	86.24	75.43	76.09	79.32	87.00	82.83	84.91	62.71	87.76	72.39	**88.52**	85.58
12	44.48	60.04	49.39	76.56	66.28	70.61	77.23	52.55	75.50	**81.65**	73.78	80.60
13	28.42	49.47	61.40	69.47	**90.18**	69.12	50.88	65.26	69.47	76.49	71.23	69.82
14	97.57	98.79	99.60	99.19	90.69	98.79	98.38	99.19	98.79	99.19	**100.00**	97.17
15	98.10	97.46	97.67	98.10	77.80	95.98	98.52	**98.31**	98.10	94.93	97.89	98.10
OA (%)	77.30	76.91	77.48	80.04	83.72	83.23	81.71	77.00	84.41	86.44	85.86	**88.55**
AA (%)	78.28	78.99	80.35	82.74	84.35	85.04	83.09	80.12	86.10	87.41	87.31	**88.89**
κ (%)	0.7538	0.7494	0.7564	0.7835	0.8231	0.8183	0.8018	0.7510	0.8311	0.8529	0.8467	**0.8757**

networks, then the transformer-based models, and finally the best results are obtained by the proposed MAEST. However, some differences between the dataset results can be observed.

For the Indian Pines dataset case, the overall worst results are obtained by KNN and RF, with the SVM being the only conventional classifier to obtain competitive results. Among the DL networks, a similar behavior can be seen, where good results are obtained for some of the classes, but lower accuracy for others. These models seem to have identified properly some classes, but were not able to generalize enough for other ones. Furthermore, even though ViT obtained a balanced accuracy, it is unable to properly model and learn the pixel spectral signatures to obtain high accuracy for many categories. However, results over an 80% of AA are only seen by SVM, 2D-CNN, both SF and MAEST versions, with the patch SF, pixel MAEST, and patch MAEST obtaining the best results in that order. The improvement brought by the specialized HS techniques introduced by SF and used in MAEST can be seen even in the pixel versions of these methods, as those obtain overall a higher accuracy than 2D-CNN without the use of information. A similar case happens comparing the pixel MAEST and the patch SF. Though SF can model many local spectral discrepancies, the robust features obtained in MAEST through reconstruction and its generalization ability make it able to outperform in almost every class of the other methods, with also good results in the rest of the classes. These results are even more notable in this specific dataset, which contains the lowest number of samples.

In the Pavia University and Houston datasets, the gap between accuracy results for different methods is not as huge. Nevertheless, a similar tendency can be seen, with the conventional classifiers behind the classic backbone networks, followed by the transformer-based models. We attribute this difference to the Pavia University and Houston datasets having more training data, and being more balanced than the Indian Pines dataset. The MAEST architecture has been specially designed to deal with this lack of information issues in HS imagery, so it is expected that the improvement difference drops in these cases. Still, the patch MAEST is able to obtain the best general results among the studied methods, with the pixel MAEST close showing the generalization and learning skills of the proposed architecture.

2) Qualitative Results: In addition to the numerical results, we also show the classification maps obtained by each compared method, including visualization of the training and testing data. The Indian Pines, Pavia University, and Houston2013 datasets in Figs. 6–8 are shown, respectively. The qualitative results follow the previously presented numerical values, with the conventional classifiers containing high levels of noise and diffuse bodies. In comparison, the DL models are able to define some edges and shapes which are better identifiable. Still, except for 2D-CNN, a high quantity of salt pepper noise distorts many shapes. Visually, it can be seen how the ViT performs in a similar way, with lots of noise distributed along the image. Both the pixel MAEST and SF show a similar pattern to ViT, but with less noise, as was expected by their similar architectures showing the improvements from the HS image specialization techniques used. Finally, patch SF and specially patch MAEST show a higher skill in defining smooth zones and building shapes with reasonable sharp edges. In the three datasets, it can be seen in the zoomed detail how MAEST is able to properly define these shapes with also more uniform categorization and texture.

D. Model Analysis

To evaluate in depth the MAEST performance, robustness, and parameters' selection, we performed a number of experiments. First, we analyze the number of spectral bands used in GSE and MGSE. Then, we discuss the difference between pixel and patch MAEST evaluating different patch

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TABLE VII

ANALYSIS OF THE NUMBER OF GROUPED SPECTRAL BANDS IN MAEST (MGSE AND GSE) IN THE INDIAN PINES DATASET IN TERMS OF OA, AA, AND κ

MAEST	Metric	Neighboring bands				
		1	3	5	7	9
Pixel	OA(%)	71.40	**78.52**	77.71	78.22	74.23
	AA(%)	78.06	**86.71**	85.15	83.77	84.35
	κ	0.6778	**0.7567**	0.7478	0.7528	0.7113
Patch	OA(%)	82.69	**84.15**	82.31	75.82	75.77
	AA(%)	90.15	**90.97**	89.46	86.67	85.54
	κ	0.8031	**0.820**	0.7991	0.7282	0.7279

TABLE VIII

ANALYSIS OF DIFFERENT VALUES OF PATCH SIZE FOR MAEST IN THE INDIAN PINES DATASET IN TERMS OF OA, AA, AND κ

Metric	Patch size				
	1	3	5	7	9
OA(%)	78.52	81.97	82.44	**84.15**	81.08
AA(%)	86.71	90.07	89.11	**90.97**	88.78
κ	0.7567	0.7953	0.8008	**0.820**	0.7852

TABLE IX

ACCURACY ON THE INDIAN PINES DATASET WITH DIFFERENT PERCENTAGES OF MASKED DATA IN THE RECONSTRUCTION PIPELINE

Masking %	Pixel AA	Patch AA
0%	82.30	87.37
25%	84.18	89.01
50%	85.32	90.23
75%	**86.71**	**90.97**

values. Next, we analyze MAEST performance by modifying the percentage of masked data and the number of epochs during reconstruction and learning rate. Finally, we compare the classification results of the proposed method with the other transformer-based state-of-the-art models in extreme situations of the quantity of training data and number of epochs.

1) MGSE and GSE Number of Bands Analysis: The first parameter studied is the number of spectral neighboring bands in the MGSE and GSE modules. Both these modules, MGSE in the reconstruction pipeline and GSE in the classification pipeline, allow MAEST to learn spectral features. Table VII shows the HS classification accuracy of the pixel and patch MAEST for the Indian Pines dataset with different numbers of neighboring bands. As shown in the table, the optimal number of bands for both the versions of the proposed MAEST is 3 bands. This difference is further appreciable in the pixel version. With only one spectral band, MAEST is unable to take proper advantage of the spectral information. However, with a higher number of neighboring grouped bands, the accuracy drops as well. We consider this relationship of lower accuracy with a higher number of grouped bands as a result of a higher spectral noise in each group of bands, which hinders learning the relationship between bands and extracting relevant features.

2) Pixel Versus Patch Analysis: As previously stated, in this work we propose two versions of MAEST: one which only uses the pixel spectral information and another one which uses the spectral–spatial contextual information surrounding the pixel as well. We defined those as MAEST pixel and patch variations for simplicity. Even though patch MAEST requires higher computational resources, it is able to obtain significantly better results than the nonspatial version over all the experimentation. Analyzing the classification results for the pixel and patch versions of both, the proposed MAEST and SF, it is evident that the use of local contextual spatial information is able to grasp useful semantic spatial relationships. The performance of 2D-CNN compared with other DL methods also suggests the importance of the use of contiguous information. Even in the visual results (Figs. 6–8), a difference along different datasets can be observed, where 2D-CNN and patch MAEST and SF barely contain noise compared with the 1-D models and the pixel versions. For further analysis, we evaluate the classification accuracy in Indian Pines with different patch sizes in Table VIII. In this table, we can observe how there is a difference of almost 4% of accuracy from the pixel MAEST to use a small patch size of 3×3. The accuracy obtained for different patch size values is similar between 3×3 and 7×7, with the last one obtaining the best results. For higher values, the patch size contains too much information and the accuracy starts decreasing as expected.

3) Reconstruction Masking Analysis: Table IX includes different AA classification results for the Indian Pines dataset, depending on the percentage of masked data in the reconstruction pipeline. The table shows how increasing the masking ratio up to a pretty high value of 75% of masked data achieves the best classification results. These high masking results are aligned with the work of He et al. [55], in contrast to other masked transformer-based models as bidirectional encoder representations from transformers (BERT) [44], whose typical masking ratio is 15% and other related computer ViTs as [30], [54] (from 20% to 50% masking ratio). Nonetheless, it is interesting to note that the major jump in accuracy happens from not using masking at all to 25%, and it increases much slower from there up to around 4% accuracy improvement. Moreover, patch MAEST without masking is able to obtain better results than patch SF and other tested methods, proving the usefulness of using the reconstruction pipeline even without masking. We link this behavior to the significant amount of spectral noise found in HS images. A high percentage of masking makes MAEST able to learn general feature representations robust to this spectral noise.

4) Learning Rate Analysis: For the proposed MAEST and many other DL networks, one of the most important hyperparameters is the learning rate. Specially in MAEST, as it contains two different pipelines which can be trained separately, a learning rate study is essential. Table X shows an AA analysis for different learning rate values in the Indian Pines dataset, for the reconstruction pipeline (R), the classification pipeline (C), and in both. In the case of the separated pipelines' analysis, we used a value of $5 * 10^{-4}$ in the other pipeline, as it is the learning rate which obtained the best results training both the pipelines (R + C) with the same learning

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TABLE X

ACCURACY (AA) ON THE INDIAN PINES DATASET WITH DIFFERENT LEARNING RATE VALUES FOR RECONSTRUCTION (R), CLASSIFICATION (C), AND BOTH (R + C)

MAEST	Learning rate				
	10^{-3}	$5*10^{-3}$	10^{-4}	$5*10^{-4}$	10^{-5}
R	88.40	84.09	87.03	**90.97**	83.77
C	88.68	76.87	88.09	**90.97**	46.04
R+C	84.88	78.15	85.16	**90.97**	36.95

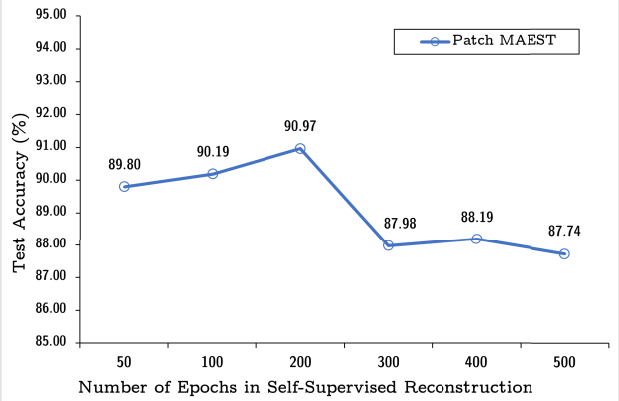


Fig. 9. Classification accuracy for the Indian Pines dataset with different epochs in reconstruction (75% masking).

rate. For the R + C analysis, it can be seen how the best learning rate values are obtained in the 10^{-4} order, with a high difference to other values. Besides, in the separated pipelines' analysis, two main results can be pointed out. First, changes in the learning rate affect more the C pipeline than R, as the classification main task is much more complex than masked reconstruction. Second, modifying the learning rate only in one of the pipelines generates lower changes in AA, reinforcing the conclusion of $5*10^{-4}$ being the best learning rate value for this case.

5) Reconstruction Epochs Analysis: Even though the fact that MAEST containing two pipelines could seem to make it converge slower than other methods, the number of epochs needed to converge the reconstruction is quite low, and once the robust representations are learned, in the classification process the performance is speeded up as well. In Fig. 9, we provide the classification accuracy results for the Indian Pines dataset depending on the number of epochs in the self-supervised reconstruction. We observe that even with only 50 epochs, the improvement in accuracy is significant, rising up to 200 epochs, and then going down again stabilizing around 88% of accuracy. We consider the reason behind this behavior is because the model over-fits and becomes unable to generalize enough for the later classification.

6) Classification Epochs' Analysis: In addition, after fixing the reconstruction epochs to 200, we compare the performance of patch MAEST, patch SF, and ViT with different numbers of classification training epochs in Fig. 10. We trained the three models from 5 epochs to 300 in the Indian Pines dataset. As expected, both SF and MAEST have a considerable

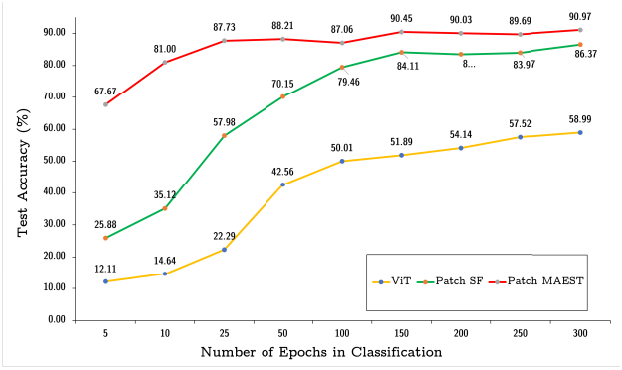


Fig. 10. Test accuracy on the Indian Pines dataset for different training epochs.

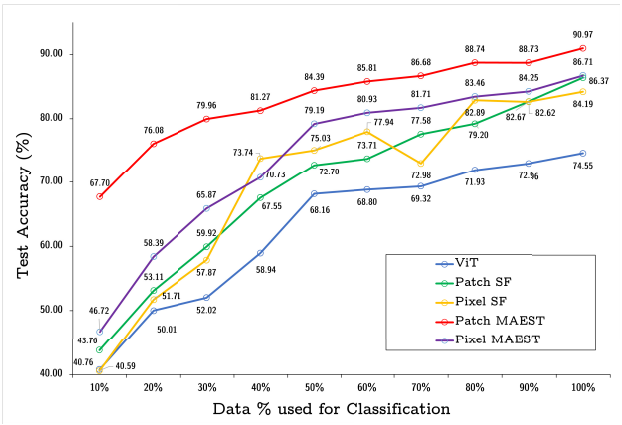


Fig. 11. Test accuracy on the Indian Pines dataset with different % of data in classification training.

difference in accuracy with the original ViT. However, it is interesting to see how the three different models behave in the first 100 epochs. It is clear how MAEST is able to perform with more than 80% of accuracy with only 10 training epochs, while the other methods are far below 50%. The proposed MAEST not only converges faster but also is able to obtain over 90% accuracy in 150 epochs, obtaining 4% accuracy improvement in almost the same total training epochs than the patch SF, thanks to the previous learning from the reconstruction.

7) Classification Data Analysis: Finally, we also carried out an experiment to analyze the effect of the amount of training data we trained from 10% of the original data, in jumps of 1/10 up to the complete training dataset of Indian Pines. The results for ViT, pixel and patch SF, and pixel and patch MAEST are shown in Fig. 11. In the graph, we observe how ViT has overall worse results than other transformer-based models, as was expected due to them being designed for HS image classification. It can also be notes how the pixel SF has quite unstable results depending on the percentage of training data used, while the patch SF and both the MAEST models steadily increase accuracy with more training data. While the patch SF and the pixel MAEST have a small fix difference in their accuracy for any percentage of data used, a huge gap can be seen between them and the patch MAEST, especially with

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lower data percentages. The spectral–spatial robust feature representations learned in the reconstruction pipeline allow the patch MAEST to have almost 70% accuracy with only 1/10th of the already limited Indian Pines training set and a much steady increase in accuracy with more data.

V. Conclusion

This article proposed a novel MAEST specially designed to classify HS remote sensing images. Unlike other existing models, MAEST takes advantage of two collaborative branches to alleviate intrinsic noise inside the network topology. Whereas the first path adopts a masking auto-encoding strategy to uncover the most robust encoding features, the second transformer path pursues to learn how to classify the complete input data while exploiting these robust features. In this way, more accurate predictions and faster convergences can be achieved. The conducted experiments showed competitive performance of the proposed approach with respect to multiple state-of-the-art classification models.

One of the first conclusions that arises from this work is the importance of transformer-based technologies in the context of HS image classification. In contrast to conventional classifiers or classic backbone networks, transformers are able to provide superior abilities in characterizing sequential spectral data which may eventually make the difference within the HS image domain in terms of performance. However, it is important to note that not all the tested transformers achieved the same positive results. In this regard, another important conclusion is related to the understanding of the HS field for successful exploitation of this technology. Being noise one of the main intricacies of HS data, learning more robust feature patterns within the transformer encoder becomes an outstanding strategy as the proposed model showed. Even though the obtained results are certainly promising, there are still important challenges for future research based on extending this work to multimodal and multitemporal data.

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